

# Universidad Gerardo Barrios

Facultad de Ciencia Y Tecnología.

Ingeniería en Sistemas y Redes de Informática.



Universidad  
Gerardo Barrios

## **Desarrollo de aplicaciones con Inteligencia Artificial – Implementación y Escalabilidad**

Nombre del proyecto

**Herramienta Autónoma para la Gestión de Ahorros y Presupuestos  
Personales**

Nombre de la Aplicación

**Ex-Codox**

Presentado por:

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## Evidencia de ejecución local

Las verificaciones se ejecutaron sobre el código real del repositorio, no sobre un proyecto de demostración.

### Expo

Comando:

```
cd ex-codox0
```

```
npm test
```

### Resultado observado:

```
(.venv) PS C:\Users\lunaa\OneDrive\Escritorio\EX-CODOX0> npm test

> expense-tracker-ai@1.0.0 test
> node --test tests/*.test.js

✓ el repositorio cliente conserva el contrato de endpoints financieros (1.1947ms)
✓ el cliente adjunta el token Firebase y no contiene claves privadas de IA (0.8532ms)
✓ el ejemplo de entorno documenta API y Firebase sin valores reales (0.4499ms)
i tests 3
i suites 0
i pass 3
i fail 0
i cancelled 0
i skipped 0
i todo 0
i duration_ms 79.1503
```

### FastAPI

Comando:

```
cd Finance_AI_API
```

```
.\.venv\Scripts\python.exe -m unittest discover -s tests -v
```

```

(.venv) PS C:\Users\lunaa\OneDrive\Escritorio\EX-CODOX0> cd .\Finance_AI_API\
(.venv) PS C:\Users\lunaa\OneDrive\Escritorio\EX-CODOX0\Finance_AI_API> .\venv\Scripts\python.exe -m unittest discover -s tests -v
C:\Users\lunaa\OneDrive\Escritorio\EX-CODOX0\Finance_AI_API\venv\Lib\site-packages\fastapi\testclient.py:1: StarletteDeprecationWarning: Using
'htpx' with 'starlette.testclient' is deprecated; install 'httpx2' instead.
  from starlette.testclient import TestClient as TestClient # noqa
test_ai_reuses_cache_and_saves_new_recommendations (test_api_contracts.ApiContractTests.test_ai_reuses_cache_and_saves_new_recommendations) .
... ok
test_all_ai_capabilities_return_the_expected_contract (test_api_contracts.ApiContractTests.test_all_ai_capabilities_return_the_expected_contr
act) ... 2026-08-06 16:46:51,409 | INFO | HTTP Request: POST http://testserver/ai/summary/test-user-uid-1234567890 "HTTP/1.1 200 OK"
2026-08-06 16:46:51,413 | INFO | HTTP Request: POST http://testserver/ai/analyze/test-user-uid-1234567890 "HTTP/1.1 200 OK"
2026-08-06 16:46:51,416 | INFO | HTTP Request: POST http://testserver/ai/recommend/test-user-uid-1234567890 "HTTP/1.1 200 OK"
2026-08-06 16:46:51,420 | INFO | HTTP Request: POST http://testserver/ai/predict/test-user-uid-1234567890 "HTTP/1.1 200 OK"
2026-08-06 16:46:51,424 | INFO | HTTP Request: POST http://testserver/ai/classify/test-user-uid-1234567890 "HTTP/1.1 200 OK"
ok
test_chat_is_scoped_to_the_authenticated_user (test_api_contracts.ApiContractTests.test_chat_is_scoped_to_the_authenticated_user) ... 2026-08-
-06 16:46:51,427 | INFO | HTTP Request: POST http://testserver/ai/chat "HTTP/1.1 200 OK"
2026-08-06 16:46:51,430 | INFO | HTTP Request: POST http://testserver/ai/chat "HTTP/1.1 403 Forbidden"
ok
test_financial_data_is_served_through_the_authenticated_api (test_api_contracts.ApiContractTests.test_financial_data_is_served_through_the_au
thenticated_api) ... 2026-08-06 16:46:51,467 | INFO | HTTP Request: GET http://testserver/data/financial/test-user-uid-1234567890 "HTTP/1.1 2
00 OK"
ok
test_ocr_accepts_an_image_and_returns_transaction_fields (test_api_contracts.ApiContractTests.test_ocr_accepts_an_image_and_returns_transacti
on_fields) ... 2026-08-06 16:46:51,471 | INFO | HTTP Request: POST http://testserver/ai/ocr "HTTP/1.1 200 OK"
ok
test_ocr_normalizes_only_valid_receipt_dates (test_api_contracts.ApiContractTests.test_ocr_normalizes_only_valid_receipt_dates) ... ok
test_ocr_parser_ignores_model_text_after_the_first_json_object (test_api_contracts.ApiContractTests.test_ocr_parser_ignores_model_text_after
the_first_json_object) ... ok

```

Los casos cubren las cinco capacidades de IA, aislamiento por usuario, autenticación 401/403, chat, OCR y fechas del recibo, datos financieros, recordatorios, caché y persistencia de recomendaciones.

## Pipeline CI/CD

Se agregó `.github/workflows/ci.yml` con dos trabajos y sin secretos reales:

- mobile: npm ci, pruebas, TypeScript y ESLint.
- api: instalación de requisitos, pruebas unitarias y compileall.

The screenshot shows the GitHub Actions interface for a workflow named "evitar dependencia de firebase-admin.json en CI #5". The workflow is in a "Success" state, triggered by a push to the "main" branch by user "LunaAcosta" 17 minutes ago. The total duration of the workflow is 1m 26s.

The workflow consists of two jobs:

- Expo checks**: Duration 1m 22s.
- FastAPI tests**: Duration 30s.

The interface includes a sidebar with navigation options: Summary, All jobs, Run details, Usage, and Workflow file. The main content area shows the workflow summary and a detailed view of the "ci.yml" file, which is triggered on push.